01 / THE NEW RESULT

Clock-Relation Reconstruction

Time as a comparison record; four readings from one calibrated scan

A known relation between numerator and denominator changes can supply the missing observation direction. The decisive quantity is the relation between their *relative changes*. Merely attaching timestamps to unchanged readings adds no information.

This note extends the preceding scale-compass protocol. Keep the positive native parameters q, p and its normalized loop invariant

$$\chi(q,p) = \frac{16q^2p^2}{(1+q^2)^2(1+p^2)^2}.$$

The clock-free algebra is unchanged. An optional physical adapter uses photon energy and vacuum flight length to express the two inputs through time comparisons. Independently, a native observer theorem determines when a single known coupled scan is enough to recover both unknowns.

Main development. Let each scan step multiply length by a known $s > 0$ and source energy by a known $t > 0$, with the transport calibration fixed. If

$$\boxed{s \neq 1, \quad t \neq 1, \quad s \neq t, \quad st \neq 1,}$$

four normalized scalar readings uniquely recover positive q, p, subject to explicit consistency checks. A fifth reading is predicted without another fit. The sufficient observation operator and its determinant are derived below.

A concrete route through time readings

For a specified photon energy $E = \hbar\omega$, put $\Theta = 1/\omega$. For a calibrated vacuum flight path put $\Delta = \ell/c$. The inherited proposed transport dictionary becomes

$$\boxed{q = \alpha\frac{\Delta}{\Theta}, \qquad p = \frac{\zeta}{\Delta\Theta}, \qquad \zeta := \beta\frac{\hbar G}{c^5}.}$$

Consequently, once q, p have been reconstructed,

$$\alpha = q\frac{\Theta}{\Delta}, \qquad \zeta = p\Delta\Theta.$$

The first calibration uses only a dimensionless duration ratio and survives a common clock gain. The second is a retained time-squared record; it does not separately determine β and G.

New result	A general comparison matrix, a four-reading scan theorem, a fifth-reading recurrence, and a complete rank classification for positive fixed-step scans.
Inherited foundations	Native cut arithmetic, Cayley transports, the scale-compass invariant and its stationary response normalization.
Physical imports	Photon energy–frequency and calibrated vacuum flight relations only. The native transport-to-source coupling remains a proposed adapter.
Exact verification	7 symbolic certificates, 7 replays, 299 checks, 15 negative controls and 28 synthetic scans.

02 / THE PHYSICAL DICTIONARY

Turn known time relations into records

Use the preceding physical dictionary

$$q = \alpha \frac{E\ell}{\hbar c}, \qquad p = \beta \frac{GE}{c^4 \ell}, \qquad \alpha, \beta > 0.$$

Here E is the source-energy scale specified by that adapter, not a graph-cost eigenvalue. The symbol α denotes a calibration factor, not the fine-structure constant.

The external inputs are clearly delimited:

Input	Relation used	Operational meaning
Photon energy [1]	$E = \hbar\omega$	Energy of the specified photon; ω is angular frequency.
Vacuum path [2]	$\ell = c\Delta$	Calibrated light-travel duration for the path, in the chosen comparison frame.
Phase-time definition	$\Theta = 1/\omega$	Inverse angular frequency, or an oscillation period divided by 2π.

The references [1–2] establish the physical input relations, not their proposed coupling to q, p. For round-trip ranging, Δ is half the round-trip delay after known instrumental offsets are removed. For multiple clock frames, the conversion between their readings must be retained.

TR1. Delay–phase-time translation. Substituting $E = \hbar/\Theta$ and $\ell = c\Delta$ gives

$$q = \alpha \frac{(\hbar/\Theta)c\Delta}{\hbar c} = \alpha \frac{\Delta}{\Theta}, \qquad p = \beta \frac{G\hbar}{c^5 \Delta \Theta} = \frac{\zeta}{\Delta\Theta}.$$

This is cancellation in the already declared adapter. It introduces no physical time evolution into the native cell.

For known positive duration ratios

$$a = \Delta'/\Delta, \qquad b = \Theta'/\Theta,$$

with fixed α, ζ, the new parameters are

$$\boxed{q' = \frac{a}{b} q, \qquad p' = \frac{1}{ab} p.}$$

Thus the previous length/energy ratios are $s = a$ and $t = b^{-1}$. Put $r = q/p$, $k = qp$ and $Y = 4/\sqrt{\chi}$. The response becomes

$$\boxed{Y(a, b) = ra^2 + \frac{1}{ra^2} + \frac{k}{b^2} + \frac{b^2}{k}.}$$

Its denominator-cleared identity is certified by the native compiler.

Retain the source convention

For N equal-frequency photons, total energy is $N\hbar\omega$. A fixed known N can be included explicitly. A fixed but unknown N still permits energy-ratio comparisons, but the formula above then recovers $N\alpha$, not α separately. Changing N during a scan changes the proposed input law and must be recorded.

Θ is not an uncertainty duration and is not the elapsed laboratory waiting time. In particular, no energy–time uncertainty inequality has been replaced by an equality. The inherited GE2 history ledger also remains distinct from these calibrated physical duration readings.

03 / GENERAL OBSERVABILITY

Keep every numerator and denominator

Suppose $q = N_q/D_q$ and $p = N_p/D_p$. At setting j, let the known relative changes be $n_{qj}, d_{qj}, n_{pj}, d_{pj} > 0$. Their net effects are

$$u_j = \frac{n_{qj}}{d_{qj}}, \qquad v_j = \frac{n_{pj}}{d_{pj}}, \qquad q_j = u_j q, \quad p_j = v_j p.$$

The values of q, p may be unknown even when these relative changes are calibrated. Equal numerator and denominator changes cancel inside their own record; unequal changes remain visible.

TR2. Four-comparison reconstruction. Let

$$\mathbf{c} = \begin{pmatrix} r \\ r^{-1} \\ k \\ k^{-1} \end{pmatrix}, \qquad r = q/p, \quad k = qp,$$

and construct the known matrix with four rows

$$\mathcal{O}_{j,:} = \left(\frac{u_j}{v_j}, \frac{v_j}{u_j}, u_j v_j, \frac{1}{u_j v_j} \right).$$

Then $\mathbf{Y} = \mathcal{O}\mathbf{c}$, where $Y_j = 4/\sqrt{\chi(q_j, p_j)}$. If $\det \mathcal{O} \neq 0$, compute

$$\widehat{\mathbf{c}} = \mathcal{O}^{-1}\mathbf{Y}.$$

The four readings belong to exactly one positive pair (q, p) under these changes if and only if

$$\boxed{\widehat{c}_i > 0 \text{ for every } i, \qquad \widehat{c}_1\widehat{c}_2 = 1, \qquad \widehat{c}_3\widehat{c}_4 = 1.}$$

When admitted, the pair is

$$\boxed{q = \sqrt{\widehat{c}_1\widehat{c}_3}, \qquad p = \sqrt{\widehat{c}_3/\widehat{c}_1}.}$$

Proof. Expanding the positive-root response gives

$$Y_j = q_j/p_j + p_j/q_j + q_j p_j + (q_j p_j)^{-1}.$$

Substitution yields the displayed linear observation equation. Invertibility makes its coefficient vector unique. Necessity of positivity and the reciprocal constraints is immediate. Conversely, these constraints imply the vector has the form (r, r^{-1}, k, k^{-1}) with positive r, k; the displayed positive roots reconstruct the unique q, p and reproduce all four readings. □

Any additional known setting has a predicted Y from its observation row and hence a predicted invariant $\chi = 16/Y^2$. It is not needed in the inversion.

Why a one-parameter scan can be sufficient

The four functions of a single scan coordinate can have independent sampled values. The number of experimental knobs is therefore not the rank of the observation matrix. Known unequal time relations can replace two independently moved axes.

Four-dimensional rank is a sufficient *linear* criterion. The coefficient vector obeys two nonlinear constraints, so it is not a universal lower bound on the number of scalar observations. The preceding three-reading, two-axis protocol remains valid and uses those constraints directly.

04 / A NATIVE OBSERVER

One known coupled scan

Take fixed positive relative step factors s, t. At integer setting $j \geq 0$,

$$\ell_j = s^j \ell_0, \quad E_j = t^j E_0, \qquad q_j = (st)^j q, \quad p_j = (t/s)^j p.$$

For the photon/time adapter this means $\Delta_j = s^j \Delta_0$ and $\Theta_j = t^{-j} \Theta_0$. The integer j labels settings; it is not a physical time variable.

Put $A = s^2$, $B = t^2$. The derived comparison operator and readout are

$$D = \operatorname{diag}(A, A^{-1}, B, B^{-1}), \qquad C = (1, 1, 1, 1),$$

$$\boxed{Y_j = CD^j \mathbf{c} = rA^j + r^{-1}A^{-j} + kB^j + k^{-1}B^{-j}.}$$

This four-coordinate coefficient carrier is a record of reciprocal comparisons, not a physical four-dimensional spacetime or an assumed quantum state space.

TR3. Exact rank criterion. With $\lambda = (A, A^{-1}, B, B^{-1})$, the four-setting observer is

$$\mathcal{O} = \begin{pmatrix} C \\ CD \\ CD^2 \\ CD^3 \end{pmatrix}, \qquad \det \mathcal{O} = \prod_{i<j} (\lambda_j - \lambda_i).$$

The native compiler verifies this determinant identity as a polynomial in four independent central coefficients. For positive s, t, the factors are all nonzero precisely when

$$s \neq 1, \qquad t \neq 1, \qquad s \neq t, \qquad st \neq 1.$$

TR2 therefore applies. Under this condition the successive readout row spaces have ranks $1, 2, 3, 4$: the observation retains one further linear comparison direction at each setting.

TR4. A parameter-free fifth-reading recurrence. Define

$$S = A + A^{-1} + B + B^{-1}, \qquad T = (A + A^{-1})(B + B^{-1}) + 2.$$

The reciprocal factorization

$$(z - A)(z - A^{-1})(z - B)(z - B^{-1}) = z^4 - Sz^3 + Tz^2 - Sz + 1$$

is certified after multiplying through by AB. Substitution of the diagonal D annihilates every coordinate. Multiplying by CD^j and $\mathbf{c}$ proves

$$\boxed{Y_{j+4} = SY_{j+3} - TY_{j+2} + SY_{j+1} - Y_j.}$$

The coefficients depend only on the known step factors. Neither q nor p is fitted for this prediction.

The recurrence is a necessary output law, not the complete admission test. Positive coefficient vectors that violate the two reciprocal constraints can still obey it. Both checks are retained in the executable.

05 / EXACT RECONSTRUCTION

A complete five-setting example

Choose a coupled program that doubles flight length and quadruples photon frequency at each setting:

$$s = 2, \qquad t = 4, \qquad \Delta_j = 2^j \Delta_0, \quad \Theta_j = 4^{-j} \Theta_0.$$

This is a specified control program. It is not asserted to be a spontaneous physical time law. A single control parameter $x = 2^j$ can encode it as $\Delta/\Delta_0 = x$ and $\Theta/\Theta_0 = x^{-2}$.

The comparison operator and its exact determinant are

$$D = \operatorname{diag}(4, 1/4, 16, 1/16), \qquad \det \mathcal{O} = \frac{136632825}{16384} \neq 0.$$

The known recurrence coefficients are $S = 325/16$ and $T = 4497/64$.

Observed inputs and an unused setting

Generate an exact native test cell with hidden $q = 2, p = 3$. At each setting the executable constructs the transported four-cycle, its stiffness matrix and its root Schur response. It normalizes the two response traces using the inherited formula

$$\chi_j = 2 \frac{\operatorname{tr} K_{\mathrm{eff},j}}{\operatorname{tr} K_{\mathrm{bare},j}}.$$

The common gains in the five frames are $2, 3, 4, 5, 6$; they cancel separately. The decoder receives only the first four normalized χ_j values and the known s, t.

j	Δ_j/Δ_0	Θ_j/Θ_0	Role	$Y_j = 4/\sqrt{\chi_j}$
0	1	1	Input	25/3
1	2	1/4	Input	9509/96
2	4	1/16	Input	2375825/1536
3	8	1/64	Input	605028929/24576
4	16	1/256	Held out	154685933825/393216

Native inversion yields

$$\widehat{\mathbf{c}} = (2/3, 3/2, 6, 1/6)^T, \qquad r = 2/3, \quad k = 6, \quad \boxed{q = 2,\ p = 3.}$$

The fifth-reading prediction is

$$\chi_4 = \frac{2473901162496}{23927738123312279130625},$$

which equals the independently constructed fifth native loop invariant exactly. The result is a synthetic model check, not an observation of a gravitational signal.

Coverage and implementation limits

The main grid covers all 25 pairs $q, p \in \{1/3, 1/2, 1, 2, 3\}$. Two other admissible step programs and one illustrative time-reference case bring the total to 28 scans. Every scan reconstructs both parameters and predicts its fifth reading.

The executable uses exact native rational arithmetic. Square roots are accepted only when rational and are otherwise retained as an unimplemented real-completion task rather than approximated. The general theorem uses positive real roots. No statistical noise model or conditioning guarantee is inferred from exact arithmetic; the example's large range is a transparent algebraic test, not an optimized instrument design.

06 / WHERE A CLOCK CAN BE BLIND

Classify the surviving ambiguities

For fixed positive step factors, the rank failures of TR3 have a direct meaning. The following transformations preserve every scalar reading along the corresponding scan, not merely the first four.

Step relation	Rank	Indistinguishable pair	What is invisible
$t = 1$, $s \neq 1$	3	$(q, p) \leftrightarrow (1/p, 1/q)$	Product $k \leftrightarrow k^{-1}$
$s = 1$, $t \neq 1$	3	$(q, p) \leftrightarrow (p, q)$	Ratio $r \leftrightarrow r^{-1}$
$s = t \neq 1$	2	$(q, p) \leftrightarrow (q, 1/p)$	The constant p branch
$st = 1$, $s \neq 1$	2	$(q, p) \leftrightarrow (1/q, p)$	The constant q branch
$s = t = 1$	1	A whole invariant level set	No changed setting

Proof. The first two rows follow by collecting $k + k^{-1}$ or $r + r^{-1}$ in the response. In the third row $p_j = p$; in the fourth $q_j = q$. The factor $4x^2/(1 + x^2)^2$ is unchanged under $x \mapsto 1/x$. The last row repeats one value of χ. The ranks count the distinct positive entries of D. □

These are global ambiguities on the positive parameter space. At a reciprocal fixed point such as $p = 1$, a particular branch pair can coincide. Known branch restrictions can also change an inverse problem; they must be stated rather than inferred from the scalar readings.

A physical phase must use the native composition law

The inherited Cayley transport is

$$U(q) = \frac{(1 - q^2)\mathbf{1} + 2qR}{1 + q^2}, \qquad R^2 = -\mathbf{1}.$$

Its exact composition is

$$\boxed{U(q_2)U(q_1) = U\left(\frac{q_1 + q_2}{1 - q_1 q_2}\right)}$$

when $1 - q_1 q_2 \neq 0$. Write $n = q_1 + q_2$, $d = 1 - q_1 q_2$. The three compiler certificates verify

$$\begin{aligned} d^2 + n^2 &= (1 + q_1^2)(1 + q_2^2), \\ d^2 - n^2 &= (1 - q_1^2)(1 - q_2^2) - 4q_1 q_2, \\ 2nd &= 2q_1(1 - q_2^2) + 2q_2(1 - q_1^2). \end{aligned}$$

For example, composing parameters $1/2$ and $1/3$ gives parameter 1, not their ordinary sum $5/6$.

Therefore $q = \alpha\omega\Delta$ is a proposed source-action input calibration; it does not identify $\omega\Delta$ with an additive rotation angle of U. If an experiment makes that angle identification, a phase-to-Cayley conversion is required and the fixed-step transport law must be checked again. The comparison-matrix method can then be used with the actually calibrated changes whenever its hypotheses hold.

07 / WHAT THE CLOCK REMOVES

A dimensionless calibration survives

TR5. Shared affine-clock cancellation. Let one clock report $\widetilde{t} = gt + o$ with fixed $g > 0$ and offset o. Differences of its timestamps give

$$\widetilde{\Delta} = g\Delta, \qquad \widetilde{\Theta} = g\Theta.$$

The origin cancels in each duration, and the common gain cancels in their ratio:

$$\boxed{q\frac{\widetilde{\Theta}}{\widetilde{\Delta}} = q\frac{\Theta}{\Delta} = \alpha.}$$

The native compiler certifies the cross-multiplied identity for arbitrary pairs of timestamps. The same fixed gain also cancels from all relative scan ratios. A varying clock gain or two unrelated clock gains needs its own comparison record.

The gravity-side record transforms as a time-squared quantity should:

$$p\widetilde{\Delta}\widetilde{\Theta} = g^2\zeta.$$

It needs a seconds calibration to be expressed in seconds squared. Even after that calibration, the factorization $\zeta = \beta\hbar G/c^5$ does not separately identify its factors.

An exact calibration example

Take illustrative duration references $\Delta = 3$, $\Theta = 5$ in one common unit chart and a native cell

$$q = 21/55, \qquad p = 13/255.$$

The scan reconstructs those parameters and returns

$$\boxed{\alpha = 7/11, \qquad \zeta = 13/17.}$$

Under the clock map $\widetilde{t} = 7t + 11$, durations 3 and 5 become 21 and 35. The reconstructed α stays $7/11$, while the time-squared record gains the factor 49. If only the second clock gain is doubled, the inferred α is wrong by a factor two; the negative control detects this missing scale record.

The comparison circle also has a clock form

At the nontrivial compatible cell $q = p = 1$, the proposed dictionary gives

$$\Delta_*^2 = \frac{\zeta}{\alpha}, \qquad \Theta_*^2 = \alpha\zeta.$$

For $\sigma = \Delta/\Delta_*$ and $\eta = \Theta/\Theta_*$,

$$q = \sigma/\eta, \qquad p = (\sigma\eta)^{-1}.$$

The preceding circle is recovered directly:

$$\boxed{\frac{1}{\sqrt{\chi}} - 1 = X^2 + Z^2, \qquad X = \frac{\sigma - \sigma^{-1}}{2}, \quad Z = \frac{\eta - \eta^{-1}}{2}.}$$

This is the inherited scale-compass identity with the source-energy axis inverted. Its 16 exact clock-coordinate checks are included; it is not presented as a second new circle theorem or a physical spacetime result.

The native GE2 ledger $\tau = \sum h_\gamma$ is an already constructed additive history budget. No equation here equates it with Δ, Θ, or an SI clock. Such a conversion would be another calibrated relation to retain.

08 / EVIDENCE AND NEXT TARGET

A reproducible observation theorem

Evidence	Coverage
7 symbolic certificates	Delay/phase-time response identity; four-mode determinant; reciprocal annihilator; shared affine-clock cancellation; three inherited Cayley composition identities.
7 serialized replays	Certificates are checked again in their declared native polynomial presentations.
299 exact checks	Includes compiler gates/replays, observer closure, inversion, matrix responses, fifth-reading predictions and clock-circle checks.
15 negative controls	Five ambiguous step families; invalid coefficient signs or reciprocal products; recurrence without admission; irrational-root rejection; fifth-frame drift; separate clock gains; unrecorded photon number; gravity-product ambiguity; rank-four not being a universal minimum; additive-phase misidentification.
28 synthetic scans	25 base parameter pairs, two alternative step programs and one time-reference example.

The general reconstruction and classification have written proofs. The executable implements the fixed-step scan specialization using exact rational native arithmetic; it does not implement an arbitrary noisy-data inverse solver. Matrix coordinates, determinant operations and row closure use the unchanged native engine. Positive-root uniqueness is used in the inherited real completion. No physical Hilbert space, external scalar engine or numerical eigensolver is imported.

Reproduce

Extract the source archive and run from its top-level folder:

```
python3.12 verify.py
```

Expected status: `PASS_NATIVE_CLOCK_RELATIONS`. Python 3.12 checks file/source integrity and launches Node; the native engine does the mathematical arithmetic. The replay must match `CLOCK_RELATION_RESULT.json` byte for byte. All preceding packets remain unchanged under `parent/`.

The canonical engine source remains pinned to Recognition-Kernel-Framework commit

`e8e3089745bf7dc0532d01c62a5f660f5658c6f0`.

Source pins, inherited manifests and exact result records accompany the note. The PDF was rendered and visually inspected. Rebuild it by running `pdflatex` twice on its included source.

Primary sources for the optional physical dictionary

[1] National Institute of Standards and Technology, *Kilogram: Mass and Planck's Constant*, updated 1 September 2022. Used only for the photon energy–frequency relation.

https://www.nist.gov/si-redefinition/kilogram/kilogram-mass-and-plancks-constant

[2] Bureau International des Poids et Mesures, *SI base unit: metre (m)*, accessed 14 October 2026. Used only for the vacuum length–duration relation.

https://www.bipm.org/en/si-base-units/metre

Next target. Calibrate the physical source-action or phase input to the native transports, then test unused settings with retained duration, source and detector-scale records. The theorem already states exactly which known relations make recovery possible and which leave a hidden branch. It does not require time to become a primitive of the operator framework.